Graphic Era HILL UNIVERSITY

Established by an Act of the State Legislature of Uttarakhand (Adhiniyam Sankhya 12 of 2011)

University under section 2(f) of UGC Act, 1956

SYNOPSIS REPORT

PharmaTrace: A Secure Network-Based Centralized System for Pharmaceutical Verification

Bachelor Of Technology

Computer Science & Engineering

Under the guidance of-

Dr. Saumitro Chattopadhyay

(Assistant Professor)

Submitted by-

Raman Kumar(2219412-L1)

Shivcharan Sharma (2219632-H2)

DEPARTMENT OF COMPUTER SCIENCE AND ENGINEERING

Contents

- Abstract
- Introduction
- Problem of Counterfeit Drugs
- The need for Secure Verification System
- Introducing PharmaTrace
- Background Study
- Analysis of different technologies
- Centralized vs Decentralized System
- Research gap
- Objectives
- Proposed methodology
- Backend Development
- Frontend Development
- Database Design
- Security Mechanism
- QR Code Implementation
- Expected Outcomes
- Key Deliverables
- Tools and Techniques
- Applications and future scope
- Scope for future enhancement
- References

Abstract

The proliferation of counterfeit pharmaceuticals poses a significant threat to global public health and undermines the integrity of the healthcare system. Existing supply chain mechanisms often lack the transparency and robustness required to effectively track and verify drugs from production to final consumption. This project, titled "PharmaTrace," proposes the development of a secure, network-based centralized system designed for the comprehensive verification of pharmaceutical products.

The system leverages a client-server architecture, with a backend powered by Node.js and Express.js, a MongoDB database for data persistence, and a user-friendly frontend built with React.js. The core functionality of PharmaTrace is to create a verifiable log of a medicine's journey, from the manufacturer to the pharmacy. This is achieved by registering each drug batch in a centralized database and updating its status at every point of transfer within the supply chain.

To ensure the authenticity of products, each drug package will be assigned a unique QR code. Consumers can scan this QR code using a simple web application to instantly access the drug's complete transaction history, thereby verifying its legitimacy. Security is a cornerstone of the PharmaTrace system, incorporating token-based authentication (JWT), Role-Based Access Control (RBAC), SSL/TLS encryption for data in transit, and database encryption for data at rest.

By offering a high-speed, cost-effective, and secure solution, PharmaTrace aims to empower consumers, support regulatory oversight, and provide a practical tool for pharmaceutical companies to combat the menace of counterfeit drugs. This report details the project's objectives, methodology, architecture, and expected outcomes, providing a complete blueprint for its implementation.

Introduction

The pharmaceutical industry is a cornerstone of modern healthcare, responsible for the discovery, development, manufacture, and distribution of life-saving medications. The integrity of the pharmaceutical supply chain is paramount, as it directly impacts patient safety and public health. Every day, millions of people rely on prescribed and over-the-counter drugs to treat illnesses, manage chronic conditions, and improve their quality of life. The trust placed in these products is absolute; consumers and healthcare providers expect every dose to be safe, effective, and authentic.

However, this trust is increasingly threatened by the complex and often fragmented nature of the global supply chain. A single medication can pass through numerous hands—manufacturers, distributors, wholesalers, and pharmacies—before it reaches the end consumer. Each transfer point introduces a potential vulnerability, creating opportunities for counterfeit, substandard, or diverted products to infiltrate the legitimate market.

The consequences of such security breaches are severe. Counterfeit drugs not only represent a significant economic challenge for pharmaceutical companies and governments but also pose a direct and often fatal risk to patients. These fraudulent products may contain incorrect ingredients, improper dosages, or harmful substances, leading to treatment failure, adverse reactions, and in the worst cases, death. The rise of online pharmacies and unregulated distribution channels has only exacerbated this problem, making it easier for counterfeiters to reach a global audience.

In response to these challenges, there is an urgent need for innovative solutions that can enhance the security, transparency, and traceability of the pharmaceutical supply chain. This project is born out of that necessity.

Problem of Counterfeit Drugs

The issue of counterfeit drugs is a pervasive and growing global crisis. The World Health Organization (WHO) estimates that as much as 10% of medical products circulating in low- and middle-income countries are either substandard or falsified. This illicit trade is valued at billions of dollars annually, making it a lucrative enterprise for criminal organizations.

Impact on Public Health: The most devastating impact of counterfeit drugs is on human lives. Patients who unknowingly consume these products are put at extreme risk. For instance:

- Treatment Ineffectiveness: Fake antibiotics or antiviral drugs with no active ingredients fail to treat infections, leading to prolonged illness, complications, and the development of antimicrobial resistance.
- Direct Harm: Some counterfeit medicines have been found to contain dangerous substances, such as poisons or heavy metals, which can cause severe illness or death.
- Erosion of Trust: The presence of counterfeit drugs in the market can erode public trust in healthcare systems, medical professionals, and the pharmaceutical industry. Patients may become hesitant to take medications, even legitimate ones.

Economic Consequences: The economic toll of counterfeit drugs is also substantial:

- **Loss of Revenue:** Legitimate pharmaceutical companies lose billions of dollars in revenue to counterfeiters, which can stifle research and development of new, life-saving drugs.
- **Cost to Governments:** Governments incur significant costs related to enforcement, investigation, and public health campaigns aimed at combating the counterfeit drug trade.
- **Burden on Healthcare Systems:** Treating patients who have been harmed by counterfeit drugs places an additional financial burden on already strained healthcare systems.

The scale of this problem highlights the inadequacy of traditional, paper-based tracking methods and underscores the necessity for a technological solution that can provide a reliable and accessible means of verification for every stakeholder in the supply chain, especially the end consumer.

The Need for a Secure Verification System

To effectively combat the infiltration of counterfeit drugs, the pharmaceutical supply chain requires a system that ensures end-to-end traceability and allows for easy verification at the point of consumption. An ideal system should possess the following characteristics:

- **Security:** The system must be resilient to tampering and unauthorized access. All data related to a drug's provenance and journey must be protected to ensure its integrity.
- **Transparency:** Authorized stakeholders, including manufacturers, distributors, and regulators, should have visibility into the supply chain. Most importantly, consumers must have a simple way to verify the authenticity of their medication.
- **Efficiency:** The verification process should be quick and straightforward. In a real-world setting, such as a busy pharmacy or a patient's home, a complicated or time-consuming process is impractical.
- **Scalability:** The system must be capable of handling a massive volume of data, accommodating thousands of manufacturers, millions of products, and billions of transactions.
- **Cost-Effectiveness:** To encourage widespread adoption, the solution should be affordable for pharmaceutical companies of all sizes, without imposing a significant financial burden that could be passed on to the consumer.

Traditional supply chain management systems often fall short of these requirements. They are typically siloed, with each organization maintaining its own internal records. This lack of a unified, transparent system makes it difficult to track a product's complete journey and easy for counterfeiters to introduce fraudulent products by creating fake documentation.

Therefore, a modern, technology-driven approach is essential. By creating a centralized and secure digital ledger of a drug's lifecycle, from the factory floor to the pharmacy shelf, we can build a chain of custody that is difficult to forge and easy to verify. This project is dedicated to designing and implementing such a system.

Introducing PharmaTrace

This project, "PharmaTrace: A Secure Network-Based Centralized System for Pharmaceutical Verification," is designed to directly address the challenges outlined above. Our mission is to leverage established and reliable web technologies to build an efficient, secure, and user-friendly platform for tracking and verifying pharmaceutical products.
PharmaTrace is a centralized system built on a classic client-server architecture. This design choice prioritizes performance, control, and ease of development, making it a practical and scalable solution for immediate implementation. The system works by logging a medicine's entire journey in a secure, central database. This creates a single source of truth for each product's history.
The core idea is simple yet powerful:

1. **Registration:** A manufacturer registers a new batch of medicine with our system. A unique digital identity is created for this batch and stored in the database.
2. **Tracking:** At every stage of the supply chain—as the medicine moves from the manufacturer to a distributor, then to a wholesaler, and finally to a pharmacy—the transfer is recorded as a transaction in the database. This builds a verifiable chain of custody.
3. **Verification:** Each medicine package is affixed with a **QR code** that corresponds to its unique digital identity. A consumer can scan this QR code with any smartphone. The application sends a request to our server, which retrieves the complete history of that specific product and displays it to the consumer.

This immediate feedback loop empowers consumers to become active participants in ensuring the safety of their own medication. If the scanned QR code does not return a valid history or shows inconsistencies, the product is immediately flagged as potentially counterfeit.

By focusing on a centralized model, PharmaTrace offers high-speed transaction processing and allows for straightforward data management and system updates. Our approach demonstrates that a robust and effective drug verification system can be constructed without the complexity and overhead often associated with more nascent technologies, providing a pragmatic solution to a critical real-world problem.

Background Study

Existing Pharmaceutical Tracking Systems

The challenge of securing the pharmaceutical supply chain is not new, and various technologies and methodologies have been employed over the years to address it. Understanding these existing systems is crucial for contextualizing the contributions of PharmaTrace.

1. **Serialized Barcodes (1D and 2D):** The most fundamental form of tracking involves placing a unique serial number on each product package, often encoded in a barcode. The Drug Supply Chain Security Act (DSCSA) in the United States mandates a serialized "track and trace" system.
   - How it Works: Each package receives a unique serial number. This allows for product identification at the unit level rather than just the batch level.
   - Strengths: Low cost and widespread adoption of barcode scanning technology.
   - Weaknesses: Barcodes can be easily replicated. Without a robust, interconnected verification system, a copied barcode on a counterfeit product can go undetected. The verification process is often not accessible to the end consumer.
2. **Radio-Frequency Identification (RFID):** RFID technology uses electromagnetic fields to automatically identify, and track tags attached to objects. An RFID tag contains electronically stored information.
   - How it Works: RFID tags are placed on pharmaceutical packages. RFID readers can scan multiple tags simultaneously without requiring a direct line of sight.
   - Strengths: High-speed scanning and the ability to read multiple items at once, which can improve inventory management efficiency. Tags can be more difficult to clone than simple barcodes.
   - Weaknesses: RFID tags and readers are significantly more expensive than barcodes, making unit-level tagging costly. There are also concerns about radio frequency interference and potential privacy issues. Consumer access to RFID readers is virtually non-existent.
3. **Holograms and Other Physical Anti-Counterfeiting Features:** Many pharmaceutical companies use overt and covert physical features on their packaging to deter counterfeiters. This includes holograms, color -shifting inks, and security seals.

- How it Works: These features are designed to be difficult to replicate, providing a visual cue of authenticity.
    - Strengths: Can provide a quick, visual check for authenticity.
    - Weaknesses: Determined counterfeiters can often replicate these features with sufficient accuracy to deceive consumers. They do not provide any digital tracking capabilities and offer no information about the product's supply chain journey.

While these technologies have contributed to supply chain security, they each have limitations. Most notably, they often fail to empower the consumer, who is the final and most critical checkpoint in the chain. PharmaTrace builds on the concept of serialization (via QR codes) but makes the verification process universally accessible and links it to a dynamic, transparent history of the product's journey.

Analysis of different technologies

In designing PharmaTrace, a critical decision was the choice of a centralized architecture. To justify this choice, it is essential to analyze the primary alternative: blockchain technology. Blockchain has been widely touted as a revolutionary solution for supply chain management due to its principles of decentralization and immutability.

How Blockchain works

1. A drug is given a unique hash and registered on the blockchain.
2. Each time the drug changes hands, a new transaction is recorded on the ledger, cryptographically linked to a previous one.
3. This creates a permanent and transparent record of the drug’s journey that can be audited by authorized participants.

Strengths of Blockchain

- **Immutability and Trust:** Transactions cannot be altered or deleted, which builds trust without needing a central intermediary.
- **Decentralization:** The ledger is not controlled by a single entity, reducing the risk of a single point of failure.
- **Transparency:** All authorized participants can view the same ledger, providing a transparent view of the supply chain.

Weaknesses of Blockchain and Rationale for a Centralized Approach

- **Performance:** Blockchains are too slow to handle the high volume of transactions needed.
- **Cost:** They are more expensive to set up and maintain than centralized databases.
- **Complexity:** Development and management require specialized expertise.
- **Privacy:** Data cannot be deleted, which conflicts with privacy laws like **GDPR**.
- **Governance:** Making updates or changes is a slow, consensus-driven process.

Given these factors, PharmaTrace adopts a centralized architecture as a pragmatic choice. It offers the high performance, cost-effectiveness, and developmental agility needed to build a scalable and effective solution today, while still providing robust security through proven, standard methodologies.

Centralized vs. Decentralized Systems: A Comparative Summary

The choice between a centralized and a decentralized architecture is one of the most fundamental decisions in designing a system like PharmaTrace. Both models offer distinct advantages and trade-offs. The following table provides a comparative summary to further justify our selection of a centralized approach.

Feature	Centralized System (PharmaTrace)	Decentralized System (Blockchain)
Control & Governance	Controlled by a single entity. Allows for easy updates, policy enforcement, and quick decision-making.	Control is distributed among participants. Changes often require consensus, which can be slow.
Performance	High. Transactions are simple database writes, processed instantly. Capable of handling very high throughput.	Variable. Can be slow, especially on public networks. Transaction speed is limited by block time and network consensus.
Cost	Low. Utilizes standard, affordable web hosting and database services. Development and maintenance costs are predictable.	High. Can involve transaction fees ("gas"). Setup and maintenance of nodes can be expensive.
Scalability	High. Can be scaled vertically (more powerful server) or horizontally (more servers) using standard cloud architecture.	Challenging. Scaling a blockchain network to handle millions of transactions per second is a major technical hurdle.
Data Mutability	Mutable. Data can be updated or deleted by an authorized administrator. This is useful for correcting errors or complying with regulations.	Immutable. Data, once written, cannot be altered. This ensures integrity but makes corrections impossible.
Security Model	Security relies on the central authority's ability to protect its server and database (firewalls, encryption, access control).	Security is cryptographic and distributed. Relies on the integrity of the consensus mechanism. Less prone to single-point attacks.

Trust Model	Trust is placed in the central authority that manages the system. Assumes the authority is not malicious.	Trust is distributed. Participants do not need to trust each other, only the protocol. "Trustless" environment.
Development Complexity	Low. Built using standard, widely- known web technologies (Node.js, React, MongoDB). Large talent pool available.	High. Requires specialized knowledge of blockchain protocols, smart contracts, and decentralized application development.

Conclusion of Comparison: For the specific use case of pharmaceutical verification, where speed, cost-effectiveness, and ease of management are critical for widespread adoption, a centralized system presents a more practical and immediately viable solution. While a blockchain-based system offers a higher degree of trustless immutability, the trade-offs in performance, cost, and complexity are significant.
PharmaTrace's model provides robust security and reliability through conventional means, making it an optimal choice for tackling the counterfeit drug problem effectively and efficiently.

Research Gap

The background study reveals a clear landscape of existing solutions and their respective limitations. While technologies like barcodes and RFID have been implemented, and blockchain is heavily researched, a significant research gap exists in the practical implementation and evaluation of a high-performance, secure, and consumer-accessible centralized system.
Much of the recent academic and industry focus has shifted towards blockchain, often overlooking the potential of optimizing and modernizing traditional client-server architectures for this specific problem. The prevailing assumption is that centralization is inherently less secure or transparent, but this is not always the case. A well-designed centralized system, managed by a trusted entity (such as a governmental regulatory body), can offer a level of performance and user experience that is currently difficult to achieve with decentralized technologies.

The gaps that PharmaTrace aims to address are:

1. Lack of Consumer Empowerment: Most existing systems are designed for industry stakeholders (manufacturers, distributors, regulators). They do not provide a simple, direct mechanism for the end consumer to verify a product's authenticity at the point of purchase or use. This is the most critical gap PharmaTrace fills.
2. Performance Benchmarking: There is limited research that directly compares the real-world performance (transaction speed, scalability, cost per transaction) of a modern centralized system versus a private blockchain for supply chain tracking at a massive scale. Our project will provide valuable data in this area.
3. Pragmatic and Deployable Model: While many blockchain solutions remain in the conceptual or pilot phase due to their complexity and cost, there is a need for a production-ready model that can be deployed quickly and affordably. PharmaTrace is designed from the ground up to be such a model.

By focusing on a centralized architecture, PharmaTrace does not aim to compete with blockchain on its core strengths of decentralization and censorship resistance. Instead, it aims to provide a superior solution on different metrics: speed, cost, scalability, and ease of use. It fills the gap for a practical, effective, and immediately implementable system that can make a tangible impact in the fight against counterfeit drugs.

Objectives

The primarily goal of this project is to develop a comprehensive and secure system to combat the proliferation of counterfeit pharmaceuticals. To achieve this, the project is broken down into the following key objectives:

- To Design and Develop a Secure Centralized Database:
    - To create a robust and scalable database schema using MongoDB that can efficiently store and manage data related to pharmaceutical products, manufactures, supply chain transactions, and user credentials.
- To Build a Web-Based platform for Manufacture:
    - To develop a secure portal where legitimate pharmaceutical manufactures can register their company, add new drug batches to the system, and generate unique identifiers for each product.
- To Implement a System for Supply Chain Tracking:
    - To create API endpoints that allow authorized entities (e.g., distributors, pharmacies) to log the transfer of ownership or possession of a drug, creating a clear and auditable chain of custody.
- To Develop a Consumer-Facing verification Application:
    - To build a simple, intuitive, and mobile-friendly web application that allows end consumers to scan a QR code on a drug package and instantly receive its full provenance and history from the database.
- To Ensure System Security and Data Integrity:
    - To implement multi-layered security measures, including secure user authentication and authorization using JSON Web Tokens (JWT), Role-Based Access Control (RBAC) to restrict permissions, and SSL/TLS encryption to protect all data transmitted between the client and server.
- To Evaluate System's Performance and Scalability:
    - To conduct performance testing to measure key metrics such as API response times, database query speeds, and the system's ability to handle a large volume of concurrent verification requests.

Proposed Methodology

The methodology for this project follows a structured software development lifecycle, encompassing system design, technology selection, implementation, and testing. The core of our approach is the development of a three-tier client-server application.

System Architecture

PharmaTrace is built upon a traditional Three-Tier Architecture, which is well-established software application architecture that organizes applications into three logical and physical computing tiers: the presentation tier, the application tier, and the data tier.

1. **Presentation Tier (Frontend):**
    - This is the user interface and is responsible for presenting data to the user and collecting user input.
    - It will be developed as a Single-Page Application (SPA) using React.js.
    - This tier will not contain any business logic. Its primary role is to render the UI and make API calls to the application tier.
    - It communicates with the application tier via a RESTful API over the HTTPS protocol.
2. **Application Tier (Backend):**
    - Also known as the logic tier, this is the core of the application. It contains all the business logic and coordinates the application's functions.
    - It will be developed using Node.js with the Express.js framework.
    - This tier will handle:
        i. Processing incoming HTTP requests from the presentation tier.
        ii. Executing business logic (e.g., validating user input, authenticating users, creating new drug records).
        iii. Communicating with the data tier to store and retrieve data.
        iv. Formatting the data and sending it back to the presentation tier in JSON format.
3. **Data Tier (Database):**
    - This tier is responsible for the storage and management of all application data.
    - We will use MongoDB, a NoSQL document-oriented database.

- It will store data in collections of JSON-like documents. This provides flexibility and aligns well with the data structures used in our JavaScript-based application tier.
- The application tier will interact with the database using and Object Data Modeling (ODM) library like Mongoose, which provides a straightforward, schema-based solution for modeling application data.

Workflow example (Consumer Verification):

1. **Presentation Tier:** The consumer scans a QR code using their smartphone's camera. The React.js application captures the unique ID from the QR code and makes an HTTPS GET request to the Api endpoint (e.g., /api/drugs/verify/:id).
2. **Application Tier:** The Express.js server receives the request. It validates the format of the ID. It then queries the data tier to find the drug corresponding to that ID.
3. **Data Tier:** The MongoDB database receives the query, retrieves the drug's document along with its transaction history, and returns it to the application tier.
4. **Application Tier:** The server formats the retrieved data inti a JSON object and sends it back as the HTTP response.
5. **Presentation Tier:** The React.js application receives the JSON data and dynamically renders it on the user's screen, displaying the drug's history in a user-friendly format.

This architecture ensures a clear separation of concerns, making the system easier to develop, test, maintain, and scale.

Backend Development

The backend is the engine of the PharmaTrace system. It will be built using Node.js, a JavaScript runtime environment that allows us to run JavaScript on the server-side, and Express.js, a minimal and flexible Node.js web application framework.

Key Responsibilities of the Backend:

- API Endpoint Creation: We will design and implement a RESTful API with clear and consistent endpoints for all CRUD (Create, Read, Update, Delete) operations.
- Authentication and Authorization: Secure the API endpoints to ensure that only authenticated and authorized users can perform certain actions.
- Business Logic Implementation: Enforce the rules and workflows of the system (e.g., a manufacturer can only update drugs they own).
- Data Validation: Validate all incoming data from clients to ensure it is in the correct format and meets the required criteria before it is processed or stored in the database.

Proposed API Endpoints:

The API will be versioned (e.g., /api/v1/) to allow for future updates without breaking existing clients.

- Authentication (/api/v1/auth)
  - POST /register: Register a new user (manufacturer, distributor, etc.).
  - POST /login: Authenticate a user and return a JSON Web Token (JWT).
- Users (/api/v1/users)
  - GET /me: Get the profile of the currently logged-in user.
  - PUT /me: Update the profile of the currently logged-in user.
- Drugs (/api/v1/drugs)
  - POST /: (Manufacturer only) Register a new batch of drugs.
  - GET /: Get a list of all drugs registered by the logged-in manufacturer.

        - GET /:id: Get the detailed information and history of a specific drug.
        - GET /verify/:qr_id: (Public Endpoint) Verify a drug using its unique QR code ID. This is the endpoint the consumer app will use.
    - Transactions (/api/v1/transactions)
        - POST /transfer: (Authenticated users) Log a new transaction to transfer ownership of a drug. The system will verify that the user initiating the transfer is the current owner.

Technology Stack for Backend:

- Runtime: Node.js
- Framework: Express.js
- Database Interaction: Mongoose (ODM for MongoDB)
- Authentication: JSON Web Tokens (JWT) using the jsonwebtoken library.
- Password Hashing: bcrypt.js for securely hashing user passwords before storing them.
- Data Validation: express-validator for validating and sanitizing request bodies.
- Environment Variables: dotenv to manage configuration and sensitive credentials.

The backend code will be structures using a modular pattern (e.g., separating routes, controllers, and models) to enhance maintainability and readability.

Frontend Development

The frontend of PharmaTrace is the primary interface through which users will interact with the system. It is crucial that the interface is intuitive, responsive, and provides a seamless user experience. We will use React.js, a popular JavaScript library for building user interfaces.

Why React.js?

- **Component-Based Architecture:** React allows us to break the UI down into reusable components, making the code more modular, manageable, and easier to debug.
- **Declarative Views:** We can simply describe what the UI should look like for a given state and React will efficiently update and render the right components when the data changes.
- **Rich Ecosystem:** React has a vast ecosystem of libraries and tools (e.g., for routing, state management, and UI components) that can accelerate development.
- **Single-Page Application (SPA):** We can build a fast and fluid user experience where page transitions happen without requiring a full page reload from the server.

Key Components of the Frontend:

1. **Consumer Verification Portal:**
    - A clean, simple, and mobile-first interface.
    - The main feature will be a button to activate the device's camera for QR code scanning. We will use a library like react-qr-scanner.
    - Once a QR code is scanned, the application will display the retrieved drug information in a clear and understandable timeline format, showing its journey from the manufacturer to its current location.
2. **Manufacturer Dashboard:**
    - A secure, login-protected area.
    - A form for registering new drug batches, including fields for name, dosage, manufacturing date, expiry date, etc.
    - A dashboard to view and manage all registered products.
    - Functionality to view the transaction history of their products.3.

3. **Shared Components:**
    - **Authentication Forms:** Login and registration forms for users.
    - **Navigation Bar and Layout:** Consistent layout and navigation across the application.
    - No**tification/Alert System:** To provide feedback to users (e.g., "Login successful," "Drug verified").

State Management: For managing the application's state (e.g., user authentication status, fetched data), we will start with React's built-in Hooks like useState and useContext. If the application's complexity grows, we may consider integrating a dedicated state management library like Redux or Zustand.

Styling: We will use a modern CSS framework like **Tailwind CSS** or a component library like **Material-UI** to create a professional and responsive design quickly. This ensures the application looks great and functions well on all devices, from desktops to mobile phones.

Database Design

The choice of database is critical to the performance and scalability of the system. We have selected MongoDB, a leading NoSQL database, for its flexibility, scalability, and seamless integration with our Node.js backend.

Why MongoDB?

- **Flexible Document Model:** MongoDB stores data in BSON (a binary representation of JSON) documents. This schema-less nature is ideal for our application, as it allows us to store complex data structures (like a nested array of transactions within a drug document) naturally.
- **Scalability:** MongoDB is designed to scale out horizontally, allowing us to distribute data across multiple servers as the application grows.
- **Performance:** It offers high-performance reads and writes, which is essential for our verification system that needs to provide real-time responses.
- **JavaScript-Native:** The database uses a JavaScript-based query language, making it very intuitive to work with from our Node.js backend.

Proposed Database Schema (Collections):

We will use the Mongoose library to define schemas for our models, providing a layer of data validation and structure.

1. **users Collection:**
    - name (String, required)
    - email (String, required, unique)
    - password (String, required) - Will store the hashed password.
    - role (String, required, enum: ['manufacturer', 'distributor', 'pharmacy','admin'])
    - companyName (String)
    - createdAt (Date, default: Date.now)

2. **drugs Collection:**
    - name (String, required)
    - manufacturer (ObjectId, ref: 'User', required) - A reference to the manufacturer in the users collection.
    - batchNumber (String, required)o manufacturingDate (Date, required)

- expiryDate (Date, required)
- dosage (String)
- qrIdentifier (String, required, unique) - The unique ID that will be encoded in the QR code.
- currentOwner (ObjectId, ref: 'User') - The current holder of the drug.
- status (String, enum: ['In Factory', 'In Transit', 'At Pharmacy', 'Sold'])
- transactionHistory (Array of embedded documents):
    - from (ObjectId, ref: 'User')
    - to (ObjectId, ref: 'User')
    - timestamp (Date, default: Date.now)
    - location (String)
- o createdAt (Date, default: Date.now)

This schema design is efficient for our primary use case: verification. When a consumer scans a QR code, we can perform a single query on the drugs collection using the qrIdentifier. This query will return a single document that contains all the necessary information, including the complete transaction history, without needing to perform complex joins across multiple tables as would be required in a relational database.

Security Mechanisms

Security is not an afterthought in the PharmaTrace architecture; it is a fundamental component woven into every layer of the system. Given the sensitive nature of pharmaceutical data, a multi-layered security approach is essential to protect against various threats.

1. **Authentication with JSON Web Tokens (JWT):** Authentication is the process of verifying a user's identity. We will use a token-based authentication system with JWT.
    - **How it Works:**
        1. A user logs in with their email and password.
        2. The server verifies the credentials.
        3. If successful, the server generates a JWT. This token is a digitally signed JSON object that contains user information (like their ID and role).
        4. The server sends this token back to the client.
        5. The client stores this token (e.g., in an HTTP-only cookie) and includes it in the Authorization header of every subsequent request to a protected API endpoint.
        6. The server validates the JWT on each request before processing it.
    - **Mathematical Foundation:** JWTs are signed using a secret key (with an algorithm like HMAC SHA256) or a public/private key pair (like RSA). A signature is created by hashing the token's header and payload along with the secret. This ensures that the token has not been tampered with, as any change to the payload would result in an invalid signature.

2. **Password Security:** User passwords will never be stored in plaintext. We will use the bcrypt library to hash and salt passwords before storing them in the database.
    - **How it Works:** bcrypt is a strong, adaptive hashing function. "Salting" involves adding a random string to each password before hashing it. This means that even if two users have the same password, their stored hashes will be different, preventing rainbow table attacks.

3. **Authorization with Role-Based Access Control (RBAC):** Authorization is the process of determining if an authenticated user has permission to access a specific resource or perform an action.

- **How it Works:**
    1. The user's role (e.g., 'manufacturer', 'admin') is encoded into their JWT upon login.
    2. When a user makes a request to a protected endpoint, a middleware function on the server will first validate the JWT and then check the user's role.
    3. Access will be granted or denied based on predefined rules. For example, only a user with the 'manufacturer' role will be allowed to access the POST /api/v1/drugs endpoint to register a new drug.

4. **Data Transmission Security with SSL/TLS:** All communication between the client (browser) and the server will be encrypted using **HTTPS (HTTP over SSL/TLS).**
    - **How it Works:** SSL/TLS uses asymmetric (public-key) cryptography to establish a secure session between the client and server. Once the session is established, symmetric-key cryptography is used to encrypt all transmitted data.
    - **Mathematical Foundation:** This process relies on the difficulty of factoring large numbers (for RSA) or solving discrete logarithm problems. It ensures:
        - **Confidentiality:** Prevents eavesdroppers from reading the data.
        - **Integrity:** Prevents the data from being altered in transit.
        - **Authentication:** Verifies that the client is communicating with the legitimate server.

5. **Database Security:**
    - **Encryption at Rest:** We will use the encryption features provided by our database hosting service (e.g., MongoDB Atlas) to ensure that the data stored on disk is encrypted.
    - **Access Control:** The database will be configured to only accept connections from the IP address of our application server, and it will be protected with strong credentials.

By implementing these comprehensive security measures, we aim to build a system that is resilient, trustworthy, and capable of protecting the sensitive data it is designed to manage.

QR Code Implementation

The QR code is the critical link between the physical product and its digital history in the PharmaTrace system. Its implementation must be robust and efficient.

What is a QR Code? A Quick Response (QR) code is a type of two-dimensional barcode. It can store significantly more information than a traditional 1D barcode and can be scanned quickly and easily using the camera on any modern smartphone.

1. **Generation of QR Codes:**
    - When a manufacturer registers a new drug batch in the PharmaTrace dashboard, the backend server will generate a **globally unique identifier** for that product. We can use a library like uuid to generate a Universally Unique Identifier (UUID).
    - This unique ID (e.g., f81d4fae-7dec-11d0-a765-00a0c91e6bf6) will be stored in the qrIdentifier field of the drug's document in the database.
    - The frontend dashboard will then take this ID and use a client-side JavaScript library (e.g., qrcode.react) to generate the corresponding QR code image.
    - This QR code image can then be downloaded by the manufacturer and incorporated into their product packaging design and printing process.
2. **The Data Encoded in the QR Code:** To make the verification process as smooth as possible, the QR code will not just contain the raw ID. Instead, it will encode a URL that points directly to the verification page of our web application, with the unique ID as a path parameter.

    Example Encoded URL: https://www.pharmatrace.com/verify/f81d4fae7dec-11d0-a765-00a0c91e6bf6

Why encode a URL?

- **Seamless User Experience:** When a user scans this QR code with their phone's native camera app, the phone will recognize it as a web link. The user can simply tap the notification to open the verification page directly in their web browser. This eliminates the need for them to download a special app.
- **Future-Proofing:** If we later decide to build a native mobile app, we can use deep linking to have the same URL open the app instead of the browser.

3. **Scanning and Verification Process:**
    - As described in the architecture workflow, when the user opens the URL, our React.js application will load.
    - The application will extract the unique ID from the URL's path.• It will then make an API call to our backend's public verification endpoint (GET/api/v1/drugs/verify/:id).
    - The backend retrieves the drug's history from the database and sends it back.
    - The frontend displays the information to the user.

This implementation strategy makes the entire verification process incredibly simple and accessible for a non-technical user, which is a core objective of the PharmaTrace project.

Expected Outcomes

Upon successful completion of this project, we anticipate a range of tangible outcomes that will demonstrate the viability and effectiveness of the PharmaTrace system. These outcomes are divided into the primary results of the project and the key deliverables that will be produced.

Expected Results

- **A Fully Functional and Deployed Prototype:** The primary result will be a working proof-of-concept of the PharmaTrace web application. This prototype will be deployed to a cloud hosting service (such as Heroku or AWS), making it publicly accessible for demonstration and testing purposes. It will include all the core functionalities outlined in the objectives, from manufacturer registration to consumer verification.
- **Demonstrated System Performance:** We expect the system to exhibit high performance, with API response times for verification requests averaging under 200 milliseconds. This will confirm that our choice of a centralized architecture and technology stack is capable of providing the real-time feedback necessary for a positive user experience.
- **Validated Security Model:** The implemented security measures will result in a secure system that protects user data and prevents unauthorized access. We expect to demonstrate that the combination of JWT, RBAC, password hashing, and SSL/TLS provides a robust defense against common web application vulnerabilities.
- **Positive User Experience:** The consumer-facing verification portal is expected to be highly intuitive and easy to use. The seamless "scan-and-verify" workflow should provide a clear and valuable experience for users, empowering them to check their medications with confidence. The manufacturer dashboard will provide a straightforward interface for managing product data.
- **Valuable Performance Benchmarks:** The project will yield valuable data on the performance of a centralized Node.js/MongoDB stack for a large-scale track-and-trace application. This data can serve as a benchmark for future projects and for comparison against blockchain-based alternatives.
- **A Scalable Architectural Blueprint:** The resulting architecture will serve as a blueprint for a production-ready system. We expect the design to be inherently scalable, capable of being expanded to handle millions of products and users with the addition of appropriate infrastructure resources (e.g., load balancers, database sharding).

Key Deliverables

The project will culminate in the production of several key deliverables that encapsulate the work done and the knowledge gained.

1. **Software Application (The PharmaTrace Platform):**
    - Source Code: The complete, well-documented source code for the frontend (React.js) and backend (Node.js) applications, managed in a Git repository.
    - Deployed Application: A live, publicly accessible URL for the deployed PharmaTrace prototype.
2. **Project Synopsis Report (This Document):**
    - A comprehensive report detailing every aspect of the project, including the problem statement, literature review, system design, methodology, implementation details, and conclusions. This report serves as the primary academic deliverable.
3. **Presentation Materials:**
    - A slide deck for the synopsis presentation, summarizing the key points of the project in a clear and concise format. The presentation will be designed to effectively communicate the project's goals, architecture, and significance to the evaluation committee.
4. **User Manual / System Guide:**
    - A brief guide for users of the system. This will include:
        - Instructions for manufacturers on how to register, log in, and add new products.
        - Instructions for consumers on how to scan a QR code and interpret the verification results.
5. **API Documentation:**
    - Clear documentation for the REST API developed for the backend. This will describe each endpoint, the expected request format, and the format of the response. This can be generated using tools like Swagger or Postman.

These deliverables will collectively represent the complete output of the project, providing not only a functional software solution but also the documentation and analysis necessary to understand, evaluate, and potentially extend the work in the future.

Tools and Technologies

The selection of a suitable technology stack is crucial for the successful implementation of the PharmaTrace project. Our choices are guided by the principles of performance, developer productivity, scalability, and ecosystem support. We have opted for a full-stack JavaScript approach, which allows for seamless integration between the frontend and backend.

Category	Technology/Tool	Justification
Backend Development	**Node.js**	A fast and scalable JavaScript runtime. Its event-driven, non-blocking I/O model is perfect for building high-performance, data-intensive APIs.
	Express.js	A minimal and flexible web application framework for Node.js. It simplifies the process of building robust APIs and handling HTTP requests.
Frontend Development	**React.js**	A powerful JavaScript library for building component-based user interfaces. It enables the creation of fast, dynamic, and modern SPAs.
	HTML5 & CSS3	The foundational technologies for structuring and styling web content.
Database	**MongoDB**	A leading NoSQL database that offers a flexible document model, high performance, and excellent scalability, aligning well with Node.js.
	Mongoose	An Object Data Modeling (ODM) library for MongoDB and Node.js. It simplifies database interactions and provides schema validation.
API & Authentication	**RESTful API**	A standard architectural style for designing networked applications. It is stateless, scalable, and easy to consume by clients.
	JSON Web Tokens (JWT)	A compact and self-contained standard for creating access tokens. It is ideal for securing APIs and implementing stateless authentication.
Code Management	**Git**	The industry-standard distributed version control system for tracking changes in source code during software development.

	GitHub	A web-based hosting service for Git repositories. It facilitates collaboration, code sharing, and project management.
API Testing	**Postman**	A comprehensive platform for building and using APIs. We will use it for testing our backend endpoints throughout the development process.
Deployment	**Heroku/ AWS/ Docker**	Cloud platforms and containerization tools for deploying, running, and managing the application. They provide scalability and ease of management.
Package Management	**npm/ yarn**	Package managers for the JavaScript ecosystem. They are used to install and manage project dependencies for both frontend and backend.

This technology stack represents a modern, robust, and cohesive set of tools that are well-suited to the requirements of the PharmaTrace project. The widespread adoption and strong community support for these technologies will also be beneficial throughout the development process.

Applications and Future Scope

While the primary focus of PharmaTrace is to secure the pharmaceutical supply chain, the underlying architecture and concept have broad applicability. Furthermore, the current project serves as a strong foundation upon which numerous advanced features can be built in the future.

Potential Applications

The track-and-trace methodology implemented in PharmaTrace can be readily adapted for other industries that are plagued by counterfeiting and require provenance verification.

- **Luxury Goods:** The system can be used to track high-value items like watches, handbags, and designer apparel. Consumers can verify the authenticity of a product before purchase, protecting both the brand's reputation and the consumer's investment.
- **Electronics Industry:** To combat counterfeit electronic components, which can cause device failures and security risks, manufacturers can use the system to track components like microchips and batteries through the supply chain.
- **Organic and High-Value Food Products:** For products where origin and authenticity are a mark of quality (e.g., premium olive oil, manuka honey, specific types of wine), the system can provide consumers with a verifiable history of the product's journey from the farm to the table.
- **Automotive and Aerospace Parts:** Ensuring the authenticity of critical replacement parts in the automotive and aerospace industries is crucial for safety. The system can be used to track parts and prevent the use of substandard counterfeits.
- **Legal and Official Documents:** The system could be adapted to issue and track official documents, such as university degrees or land titles, providing a secure way to verify their legitimacy and prevent forgery.

Scope for Future Enhancements

The current version of PharmaTrace lays the groundwork for a powerful platform. The following enhancements could be explored in future phases of the project to further increase its capabilities and impact:

- **Native Mobile Application:** While the initial version will be a responsive web application, developing dedicated native mobile apps provide a richer user experience. This would include features like push notifications for product recalls, offline access to scanned history.

- **Integration with IoT Devices:** Integrating **IoT sensors** into shipping containers can monitor the real-time **temperature and humidity** for sensitive medications. This data is then added to the drug's transaction history, providing an extra layer of **quality assurance**.

- **Advanced Data Analytics and AI:** The centralized database will accumulate a vast amount of supply chain data. Machine learning algorithms could be applied to this data to:

    - **Predict and Flag Anomalies:** Identify unusual patterns in the supply chain that might indicate theft, diversion, or the introduction of counterfeit products.
    - **Optimize Supply Chain Routes:** Analyze transaction data to identify bottlenecks and suggest more efficient logistics pathways.
    - **Generate Insights for Regulators:** Provide health authorities with a high-level view of drug distribution, helping them to manage inventory and respond to public health emergencies.

- **Integration with Pharmacy and E-commerce Systems:** The PharmaTrace API could be integrated with the inventory management systems of pharmacies and online retailers. This would allow for automatic verification of stock and provide consumers with an authenticity guarantee at the point of online purchase.

These future enhancements demonstrate the long-term potential of the PharmaTrace platform to evolve into a comprehensive and intelligent solution for supply chain integrity across multiple industries.

References

1. World Health Organization. (2018). WHO Global Surveillance and Monitoring System for substandard and falsified medical products. [Online]. Available: https://www.who.int/publications/i/item/9789241513425
2. FDA. (2021). Drug Supply Chain Security Act (DSCSA). U.S. Food and Drug Administration. [Online]. Available: https://www.fda.gov/drugs/drug-supply-chain-integrity/drug-supply-chain-security-act-dscsa
3. Kshetri, N. (2018). Blockchain's roles in meeting key supply chain management objectives. International Journal of Information Management, 39, 80-89.
4. Musamih, A., Salah, K., Emir, E., & Jayaraman, R. (2021). A Blockchain-Based Approach for On-Demand and Secure Sharing of Healthcare Data. IEEE Access,9, 66539-66553.
5. Haber, S., & Stornetta, W. S. (1991). How to time-stamp a digital document. In A. J. Menezes & S. A. Vanstone (Eds.), Advances in Cryptology — CRYPTO '90 Proceedings (pp. 437–455). Springer.
6. Fielding, R. T. (2000). Architectural Styles and the Design of Network-based Software Architectures. Doctoral dissertation, University of California, Irvine.
7. MongoDB, Inc. (2023). The MongoDB 6.0 Manual. [Online]. Available: https://www.mongodb.com/docs/manual/
8. Haverbeke, M. (2018). Eloquent JavaScript (3rd ed.). No Starch Press. (Provides foundational knowledge on JavaScript, Node.js, and web technologies).
9. Freeman, A. (2019). Pro Express.js: Master the MEAN Stack. Apress.
10. Banks, A., & Porcello, E. (2020). Learning React: Modern Patterns for Developing React Apps (2nd ed.). O'Reilly Media.